National University of Computer & Emerging Sciences

MT2005 – Probability and Statistics

Homework # 1(f)

Measures of Dispersion

Below is the dataset containing features like **House Size (in square feet)** and **Number of Rooms**, along with the corresponding house prices:

House Size (sq ft)	Number of Rooms	House Price (K)
1200	3	250
1500	4	320
1000	2	220
1800	4	400
1600	3	350

1. **Absolute Measures of Dispersion**
   a) Calculate the **range** for **House Size** and **House Price**.
   b) Calculate the **interquartile range** for **Number of Rooms**.
   c) Calculate the **mean deviation** for **House Size**.
   d) Calculate the **standard deviation** for **House Size**, **Number of Rooms** and **House Price**.
2. **Relative Measures of Dispersion**
   a) Calculate the **coefficient of range** for **House Size**.
   b) Calculate the **coefficient of interquartile range** for **Number of Rooms**.
   c) Calculate the **coefficient of mean deviation** for **House Size**.
   d) Calculate the **coefficient of variation (CV%)** for **House Size**, **Number of Rooms** and **House Price**.
3. **Correlation Coefficient**
   a) Calculate the **correlation coefficient** between **House Size** and **House Price**.
   b) Calculate the **correlation coefficient** between **Number of Rooms** and **House Price**.
   c) Based on these correlations, describe the relationship between these features and house price.
4. **Visualization**
   a) Use Python to plot a **scatter plot** for each pair of features:
      i) **House Size vs House Price**
      ii) **Number of Rooms vs House Price**
5. **Feature Selection**
   a) If you were to train a machine learning model to predict **House Price**, which feature (House Size or Number of Rooms) would you prioritize? Explain why, based on the dispersion measures, correlation results and the trends you observed in the scatter plots.

Python Code for Scatter Plot:

Below is the Python code to create scatter plots for **House Size vs House Price** and **Number of Rooms vs House Price**.

```
import matplotlib.pyplot as plt
import pandas as pd

# Student Roll Number (replace with your actual roll number)
roll_number = "Your Roll Number Here"

# Creating the DataFrame
data = {
    'House Size': [1200, 1500, 1000, 1800, 1600],
    'Number of Rooms': [3, 4, 2, 4, 3],
    'House Price': [250, 320, 220, 400, 350]
}

df = pd.DataFrame(data)

# Scatter plot for House Size vs House Price
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.scatter(df['House Size'], df['House Price'], color='blue')
plt.title('House Size vs House Price')
plt.xlabel('House Size (sq ft)')
plt.ylabel('House Price (K)')

# Scatter plot for Number of Rooms vs House Price
plt.subplot(1, 2, 2)
plt.scatter(df['Number of Rooms'], df['House Price'], color='green')
plt.title('Number of Rooms vs House Price')
plt.xlabel('Number of Rooms')
plt.ylabel('House Price (K)')

# Adding roll number caption at the bottom of the plot
plt.figtext(0.5, 0.01, f'Roll Number: {roll_number}', ha='center', va='center', fontsize=12, color='bla

plt.tight_layout()
plt.show()
```